

Report: Biophysical Basis for Three Distinct Dynamical Mechanisms of Action Potential Initiation

Introduction

Neurons are specialised cells in the nervous system of (almost) all animals that are excitable, meaning they can generate electrical signals called 'action potentials'. An action potential refers to the electrical signal that is generated within a neuron, specifically within the axon. The action potential is mainly driven by an inflow of fast activating, depolarizing Sodium (Na^+) ions, and an outflow of slow activating hyperpolarizing potassium (K^+) ions moving in and out of the neuron, thereby creating a voltage-based event that travels along the neuron's membrane and ends at the axon terminal.

An action potential is called an 'all-or-none' signal because it either fully occurs or does not occur at all. Whether a neuron fires an action potential depends on a dynamic interaction of membrane and current parameters. Even neurons of the same type can have different stimulus thresholds and be more or less excitable than others. The number of action potentials per second (firing rate) may also vary - if an input (stimulating) current is just above the neuron's voltage threshold, the neuron may fire slowly, whereas if the input current is very strong, the neuron may fire more frequently. However, some neurons may fire only once or in a single burst regardless of the input current strength.

Literature Review

Dynamical Systems in Neuroscience - Eugene M. Izhikevich, 2007

This e-book is a fantastic resource for all things related to computational neuroscience and from which many of the definitions and explanations in this report are sourced.

The Local Electric Changes Associated with Repetitive Action in a Non-Medullated Axon - A. L. Hodgkin, 1948

Hodgkin's initial work in analysing and classifying neuron excitability was groundbreaking and despite being published in 1948, is still one of the most widely used neuronal classification frameworks today. Prior understanding of neuron dynamics seemed to assume a simplistic view of whether a neuron fires or not, whereas Hodgkin was able to demonstrate that neuron excitability was far more complex and nuanced, and depended not only on external stimulation but rather a combination of membrane and ion current dynamics. This study acted as the basis for further, and more detailed studies which resulted in the widely used Hodgkin-Huxley model for neuronal excitability.

A Quantitative Description of Membrane Current and its Application to Conduction and Excitation in Nerve - A. L. Hodgkin and A. F. Huxley, 1952

Building on the work by Hodgkin (1948), the Hodgkin-Huxley model identified the four state variables which each control the excitability of neurons. Their work on identifying key gating variables and their impact on current activation timings and speed was foundational and improved our understanding of the complex behaviours exhibited inside neurons of various types. This complex, four dimensional model was found to be biophysically accurate in describing real neuron behaviour and as such became one of the most important models in computational neuroscience to date. Its

complexity, although necessary for biophysical realism, is often considered too complicated for computational simulations, or for fixing variables in order to estimate specific parameters values. For this, a simpler model was needed, that would allow for simulation that still captured the different excitability patterns exhibited by real neurons.

Mathematical Models of Threshold Phenomena in the Nerve Membrane, Fitzhugh (1955)

Building on the Hodgkin-Huxley work, Fitzhugh described neuron models using differential equations to represent changes in state variables over time. The study identifies three types of threshold phenomena at spike initiation, based on how system trajectories behave in phase space;

1. Singular Point Threshold Phenomenon - Involves a saddle point in phase space and is most closely associated with Hodgkin's class 1.
2. Discontinuous Threshold Phenomenon - Involves discontinuous equations and associated with Hodgkin's class 3.
3. Quasi Threshold Phenomenon - Mostly associated with class 2 but can be associated with class 3

An Active Pulse Transmission Line Simulating Nerve Axon - J. Nagumo, S. Arimoto, S. Yoshizawa, 1962

Again building on the work done by Hodgkin-Huxley, this study built an electronic circuit (an active pulse transmission) that simulated the behavior of nerve axons during stimulation. As with the Fitzhugh study, the goal of interpreting spike initiating dynamics required a reduced version of the Hodgkin-Huxley model. This practical engineering approach to simulating real neurons demonstrated a physical realisation of neuron excitability and helped validate the reduced Hodgkin-Huxley model.

Both Fitzhugh and Nagumo were inspired by the Hodgkin-Huxley model, and sought to provide a general mathematical framework for understanding excitability and thresholds in neurons. Both papers took the approach of simplifying the Hodgkin-Huxley model in order to favour simulation and interpretation over biophysical accuracy. Both papers laid the groundwork for simplified neuron models, and led to the Fitzhugh-Nagumo model, a reduced version of the Hodgkin-Huxley model that is widely used in computational neuroscience today.

Voltage oscillations in the barnacle giant muscle fiber - C Morris, H Lecar, 1981

This paper investigated whether a simple system of two voltage-dependent conductances (Ca^{2+} and K^{+}) could account for the complex oscillatory behavior observed in barnacle muscle fibers when under current stimulation. The study introduced the Morris-Lecar model, a 2-dimensional, conductance-based neuron model based on a reduced Hodgkin-Huxley model. The paper analyzed how the balance between inward (Ca^{2+}) and outward (K^{+}) currents produced enough differing behaviors that it could explain the complex spiking dynamics exhibited in the muscle fiber.

This 2-dimension conductance model, introduced by Morris-Lecar, became the basis of the Prescott et al.'s model which is the focus of this report.

Using this model, Prescott et al. aimed to describe a mechanistic explanation for the Hodgkins classes of excitability in neurons by using a unified framework of parameters, which will be further discussed throughout this report.

Primary Goal of the Prescott et al. Study

According to the Hodgkin classification, the way in which a neuron fires an action potential (that is, the pattern of their frequency-current (f - I) curves) falls into three classes:

Class 1: Neurons can start firing at a low rate when given a weak perithreshold stimulus, and their firing rate increases smoothly as the stimulus increases. Their f - I curve is therefore continuous. These neurons are often referred to as tonic spiking neurons.

Class 2: Neurons do not fire at all until the stimulus exceeds the threshold, at which point they jump to a high firing rate which they maintain whilst the stimulus continues. These neurons cannot sustain low-frequency firing rates if the stimulus is relatively weak. They therefore have a discontinuous f - I curve more akin to an on/off switch.

Class 3: Neurons do not fire repetitively under steady stimulus. They might fire only a single spike (or a short burst) at the beginning of a stimulus and then fail to re-fire even if the stimulus continues. Because they often don't produce a train of spikes, a firing rate can't be defined.

Like the Hodgkins-Huxley study, the researchers wanted to explain which intrinsic properties of neurons produced those differences in spinal lamina I neurons. The researchers aimed to computationally simulate how the ion channels and currents in a neuron (i.e. its biophysical properties) determine how the neuron encodes stimuli, which then affect its spiking pattern, and therefore show that all three excitability classes can be described by a unified framework of parameters.

By identifying the specific mechanism for spike initiation for each class, the researchers hoped to answer the as-yet unanswered question of how class 3 neurons initiate spikes without being able to fire repetitively. Since class 3 neurons have historically been thought of as unhealthy, in-depth studies of its neural mechanics have been neglected. However, healthy neurons showing this spiking pattern have been observed in the spinal cord and retina ^{(7),(8)}, and so further research is needed in this area.

In addition to simulating spike initiating mechanisms, the researchers hypothesised that the three Hodgkin's classes were not mutually exclusive, and that altering a collection of neural parameters could produce class 1, 2, or 3 excitability within one/the same neuron. Using their computational simulation of spike initiating mechanics, the researchers hoped to identify the parameters that influenced excitability class, which would then allow them to replicate these parameter changes in physical lab experiments and validate their mathematical representations. By demonstrating that small changes in a neuron's ion channel dynamics could shift its excitability class, the researchers hoped to explain how certain neurons might change behavior due to circumstance, development, drugs, or disease.

Methodology

The researchers used Hodgkin's Classes of Excitability above as a basis to categorise and explain the different reactions to electrical stimuli observed in spinal lamina I neurons. Using a variation of the Morris-Lecar 2-dimensional model as a basis to model the spike initiating dynamics in each excitability class, the researchers computationally simulated the f - I curves observed in each Hodgkin's class. The Morris-Lecar 2D model is a simplified version of the 4-dimensional Hodgkin-Huxley model, which allowed the researchers to use a forward engineering approach to test their assumptions about various system dynamics computationally, and then validate those assumptions in physical lab-experiments. The researchers aimed to keep their mathematical model as simple as possible for

computational and explanatory purposes, increasing its complexity only when it was required to better explain the outcomes observed during physical experiments.

Prescott et al. Model Analysis

The Hodgkin-Huxley conductance based model includes four state variables which each control the dynamics of the neuron membrane voltage (V), outward Potassium activation (n), inward Sodium activation (m), and inward Sodium inactivation (h). This 4-dimensional model has been found to be biophysically accurate when used on real neurons, but due to its complexity is often difficult to simulate computationally for use in isolating parameter values for spike initiation.

Hodgkin-Huxley 4-dimensional conductance model:

$$\begin{aligned}
 I &= C_m \frac{dV_m}{dt} + \bar{g}_K n^4 (V_m - V_K) + \bar{g}_{Na} m^3 h (V_m - V_{Na}) + \bar{g}_l (V_m - V_l), \\
 \frac{dn}{dt} &= \alpha_n(V_m)(1 - n) - \beta_n(V_m)n \\
 \frac{dm}{dt} &= \alpha_m(V_m)(1 - m) - \beta_m(V_m)m \\
 \frac{dh}{dt} &= \alpha_h(V_m)(1 - h) - \beta_h(V_m)h
 \end{aligned}$$

Morris-Lecar, much like Fitzhugh and Nagumo, adapted the Hodgkin-Huxley model by making the assumption that the depolarizing Ca^{2+} current is so fast activating that it can be thought of as instantaneously in steady state at all times, and thus has no associated differential equation that impacts the model. This assumption allowed Morris-Lecar to simplify their model to one that was 2-dimensional, yet still showed the varying spiking patterns, bistability, and oscillations associated with each excitability class.

Morris-Lecar 2-dimensional model:

$$\begin{aligned}
 C \frac{dV}{dt} &= I - g_L(V - V_L) - g_{Ca} M_{ss}(V - V_{Ca}) - g_K N(V - V_K) \\
 \frac{dN}{dt} &= \frac{N_{ss} - N}{\tau_N}
 \end{aligned}$$

where

$$\begin{aligned}
 M_{ss} &= \frac{1}{2} \cdot \left(1 + \tanh \left[\frac{V - V_1}{V_2} \right] \right) \\
 N_{ss} &= \frac{1}{2} \cdot \left(1 + \tanh \left[\frac{V - V_3}{V_4} \right] \right) \\
 \tau_N &= 1 / \left(\varphi \cosh \left[\frac{V - V_3}{2V_4} \right] \right)
 \end{aligned}$$

Given that the 2D Morris-Lecar based model was sufficient to display the voltage dynamics produced for all three excitability classes we will use this model in our phase plane and bifurcation analysis.

Like the Morris-Lecar model, the Prescott et al. model is of the general form:

$$C \cdot \frac{dV}{dt} = I_{stim} - I_{Na} - I_K - I_{leak}$$

$$\frac{dw}{dt} = \phi_w \cdot \frac{w_{\infty}(V) - w}{\tau_w(V)}$$

The Prescott model is given in full below, along with associated differential equations and parameters:

$$C \, dV/dt = I_{stim} - \bar{g}_{fast} m_{\infty}(V)(V - E_{Na}) - \bar{g}_{slow} w(V - E_K) - g_{leak}(V - E_{leak}) \quad (2)$$

$$dw/dt = \phi_w \frac{w_{\infty}(V) - w}{\tau_w(V)} \quad (3)$$

$$m_{\infty}(V) = 0.5 \left[1 + \tanh\left(\frac{V - \beta_m}{\gamma_m}\right) \right] \quad (4)$$

$$w_{\infty}(V) = 0.5 \left[1 + \tanh\left(\frac{V - \beta_w}{\gamma_w}\right) \right] \quad (5)$$

$$\tau_w(V) = 1 / \cosh\left(\frac{V - \beta_w}{2\gamma_w}\right) \quad (6)$$

Membrane Parameters:

C = Membrane capacitance of the neuron, fixed at $2 \, \mu\text{F}/\text{cm}^2$

V = Membrane Voltage which varies with stimulation strength, (mV)

Inward and Outward Current Parameters:

I_{stim} = Inward applied current, $\mu\text{A}/\text{cm}^2$

E_{Na} = Reversal potential for natural outward hyperpolarizing Na^+ current, fixed at $50 \, \text{mV}$

E_K = Reversal potential for natural outward hyperpolarizing K^+ current, fixed at $-100 \, \text{mV}$

E_{leak} = Reversal potential for natural leak current of the neuron, fixed at $-70 \, \text{mV}$

g_{fast} = Maximum conductance of the fast inward Na^+ current, fixed at $20 \, \text{mS}/\text{cm}^2$

g_{slow} = Maximum conductance of the slow K^+ current, fixed at $20 \, \text{mS}/\text{cm}^2$

g_{leak} = Conductance of the natural voltage leakage of the neuron, fixed at $2 \, \text{mS}/\text{cm}^2$

Gating Parameters:

m = A 'gating variable' that controls the inward current of Sodium ions

w = A 'gating variable' that controls the outward current of Potassium ions

$m_{\infty}(V)$ = The steady-state activation of the fast inward current. However, just like the Ca^{2+} activation in the Morris-Lecar study, this is assumed to be instantaneous and so has no associated differential equation that impacts the model

$w_{\infty}(V)$ = The steady-state activation of the slow outward inward current that impacts the change in w

β_m = Midpoint voltage of $m_{\infty}(V)$ curve, fixed at $-1.2 \, \text{mV}$

β_w = Midpoint voltage of $w_{\infty}(V)$ curve, (mV)

λ_m = Controls the steepness of the sigmoid curve for m_{∞} , fixed at $18 \, \text{mV}$

λ_w = Controls the steepness of the sigmoid curve for w_{∞} , fixed at $10 \, \text{mV}$

Phi_w = Rate constant for the slow activating Potassium current, fixed at 0.15

$\tau_w(V)$ = Adds a voltage dependence to the change in w over time

Key Findings

The study successfully described the three distinct dynamical mechanisms by which neurons initiate an action potential, showing that each mechanism corresponded to a Hodgkin's excitability class. By simulating spike initiation in each neuron, the researchers successfully identified and isolated the range of parameters that impact a neuron's firing pattern, and discovered that excitability class is determined by a competition between the inward current that drives the membrane voltage up and an outward current that pulls the voltage down, in addition to how fast each of these processes activate. The balance of these currents and their activation time can lead to three distinct outcomes when a neuron is stimulated, showing that excitability classes are not mutually exclusive and that a single neuron could exhibit spiking patterns across all three classes depending on the range of parameters included in the computational model.

Using a simplified 2D version of the Hodgkin-Huxley model, and changing the kinetic speed of the outward current (β_w in the model), they could switch the model neuron from class 1, class 2, or 3 excitability.

In class 1 neurons, when the neuron is just at the brink of firing, the net input current (Na^+) is slightly depolarizing such that the neuron's outward currents (Ka^+) won't counteract it – the residual polarization will then make the neuron fire. The neuron's stable resting state disappears via a slow, smooth change via a type of bifurcation known as a saddle-node on invariant circle (SNIC), causing the neuron to fire in a continuous spiking pattern. Once input current reaches a certain threshold, the neuron can fire slowly because the balance of currents still slightly favors excitation, allowing a spike to be initiated in a gradual and continuous manner.

Conversely, class 3 neurons don't fire continuously. Instead, they fire once, at the onset of stimulation and then stop. This is because their resting state is always stable despite the strength of stimulation. When a stimulus starts, a fast inward current (Na^+) momentarily overcomes the slow outward current (K^+), which is enough to push the neuron across a threshold and fire one spike. But since the stronger outward current acts more slowly, it kicks in just after the initial spike and overcomes the inward current, bringing the membrane voltage back down and preventing additional spiking. This dynamic is associated with a quasi-separatrix crossing (QSC).

In class 2 neurons, the competition between inward and outward currents is more evenly distributed compared to class 1 and 3. When a stimulus reaches threshold, the fast-activating inward current overcomes the slower outward current, triggering a spike. However, as the outward current catches up, it counteracts the depolarization and pulls the membrane potential back down. As the stimulus increases beyond threshold, this competition between currents becomes cyclical as the inward current repeatedly overcomes and is then overcome by the outward current such that repetitive spiking is produced. This results in a limit cycle that produces repeated membrane voltage oscillations indicative of a subcritical Hopf bifurcation.

They also noted that altering several different parameters produced the same effects on excitability. Specifically, a neuron could exhibit class 1 behavior either by inactivation of a slow outward current or by activation of a slow inward current. This finding was confirmed in the 2D and 3D models and further supported the conclusion that excitability was determined by a competition between inward and outward currents, and their activation speeds, rather than by the strength of a stimulus or some other predetermined physical property of the neuron.

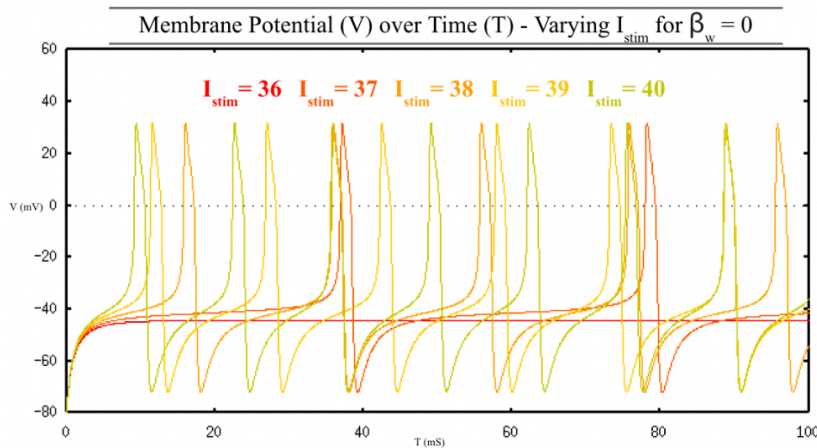
By performing phase plane and bifurcation analysis, the researchers were able to examine the system's behavior and stability at spiking threshold. The analysis showed how parameter values change at spike initiation, which produce bifurcations in system dynamics once stimuli reach perithreshold levels. By plotting nullclines for each state variable and finding their intersections they visualized what happens to system parameters the instant a neuron transitions from resting to spiking.

Time Series Analysis

To validate that our 2D Prescott model can generate the spiking patterns in each excitability class, we can plot the membrane potential (V) over time for values of I_{stim} that correspond to just before and up to the point of spike initiation for classes 1 ($\beta_w = 0$) and 2 ($\beta_w = -13$) to visualise the spike variation between classes. And for class 3 we plot values of V from just before and far beyond the point of spike initiation to show that further spiking cannot be generated when $\beta_w = -21$.

Class 1: A gradual ramp up in I_{stim} results in a gradual ramp up in firing rate. This shows the neuron's ability to fire at relatively low firing rates for weaker stimulation.

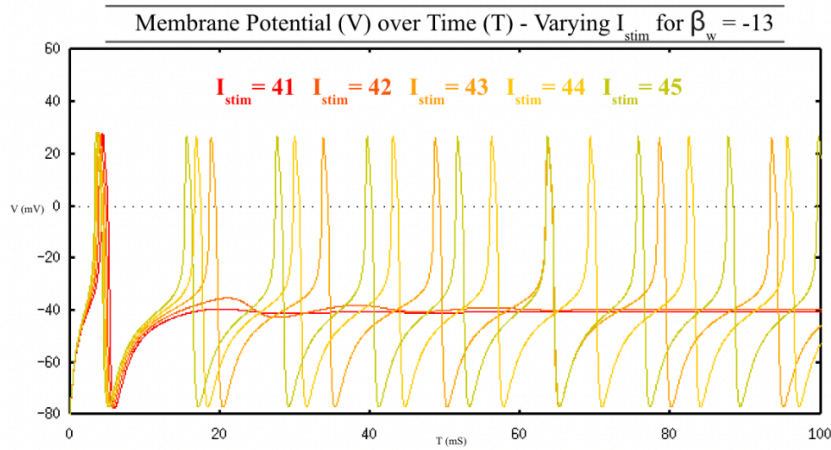
As I_{stim} crosses the critical threshold ($37\mu A/cm^2$), the stable fixed point collides with an unstable fixed point, destroying the stable fixed point through a saddle-node on invariant circle (SNIC) bifurcation. This causes the system to transition from a stable resting state to sustained periodic spiking, giving rise to a limit cycle that produces continuous oscillatory spiking behavior.



Class 2: A sudden ‘jump start’ as the fast activating inward currents win out over the slow activating outward current.

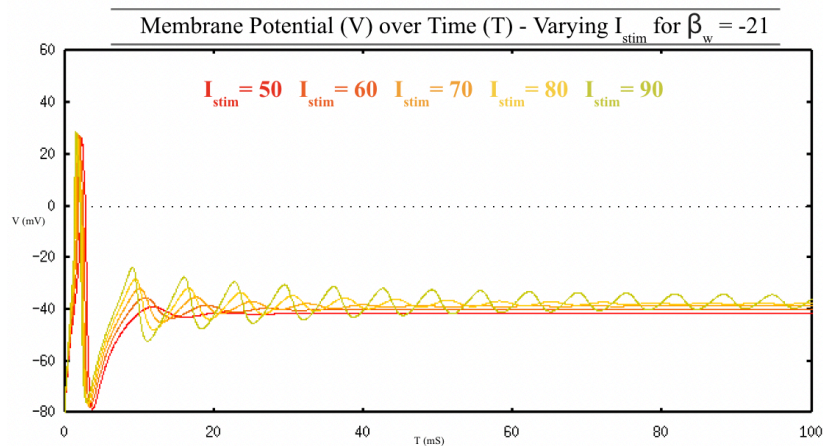
For subthreshold values ($<42\mu A/cm^2$), the stable fixed point remains intact, and the system quickly returns to its stable, resting state once the slower outward current catches up.

For I_{stim} values above threshold ($43+\mu A/cm^2$) repetitive spiking at high rate is produced. This is indicative of when the stable fixed point becomes unstable through a subcritical Hopf bifurcation, causing the system to jump into a limit cycle.

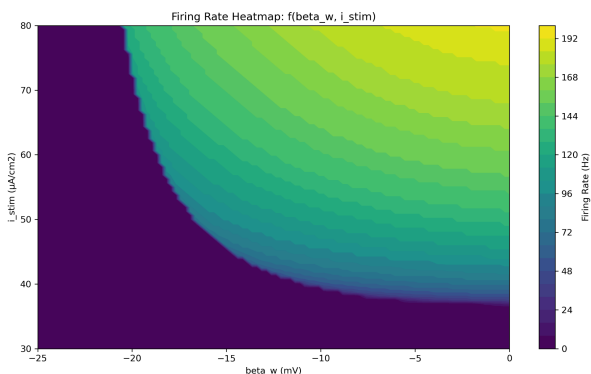


Class 3: At the onset of stimulation, an initial spike is produced when the fast activating inward current wins out over the slow activating outward current. Unlike class 1 and 2, the system remains near a stable fixed point even for high values of I_{stim} ($70+\mu A/cm^2$), since the stronger outward current causes the system to return to its stable, resting state.

The system does not undergo a bifurcation or transition into a limit cycle. This behavior reflects a quasi-separatrix crossing (QSC), where the system momentarily crosses a threshold-like boundary in phase space without losing stability.



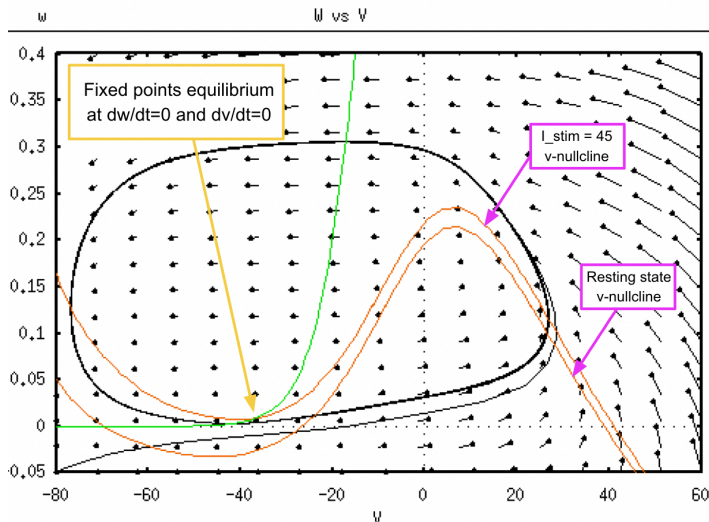
The impact of the outward and inward current competition for a class 2 Neuron can be visualised clearly in a firing rate heatmap for $\beta_w = -13$:



The inability of class 2 neurons to sustain repeated firing at relatively low (subthreshold) levels of I_{stim} or high levels of outward current ($-\beta_w$), or some combination of both is evidenced by the very clear boundary line in the heatmap.

The transition from zero firing rate (purple) to relatively high firing rate (green) is clearly visible and distinct and shows that both I_{stim} and β_w impact a neuron's transition from resting to spiking.

Phase Plane Analysis



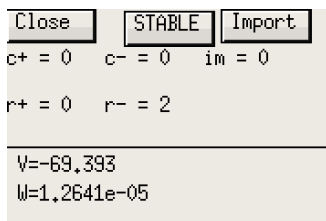
The V-nullclines (red) for the stable resting state of the neuron ($I_{stim}=0$) and after the point of excitation ($I_{stim}=45$) show a shift upward in voltage for increased I_{stim} .

The w-nullcline (green) is sigmoid shape and does not change for varying values of I_{stim} (because β_w is fixed), rising smoothly (where $dw/dt=0$)

Trajectory (black) shows the path the system takes over time.

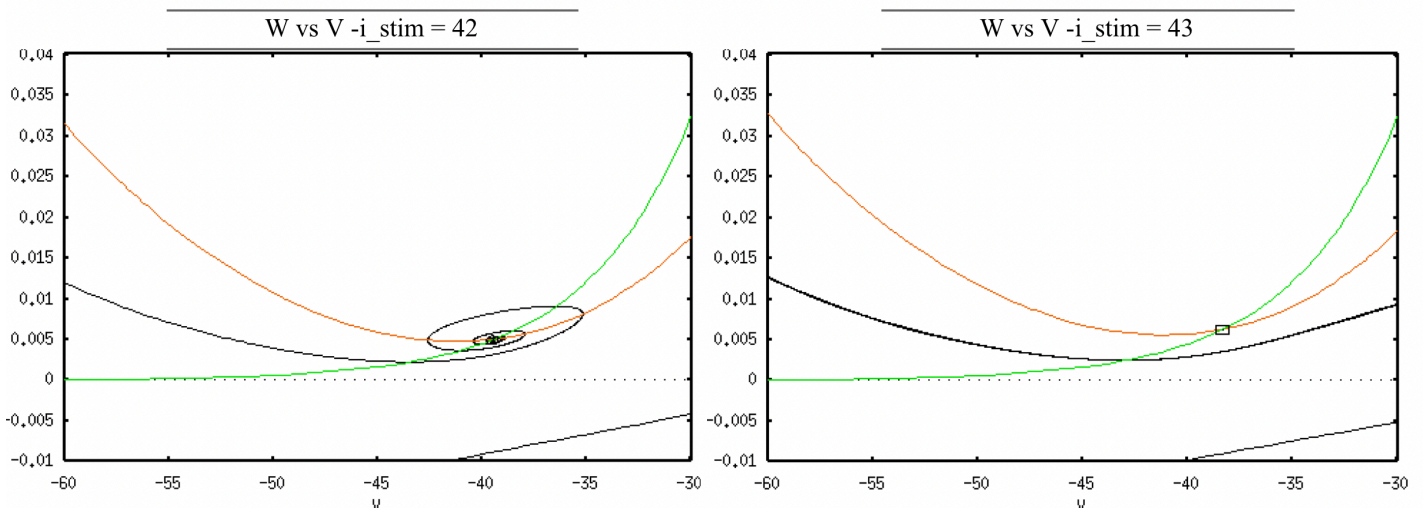
Direction field (arrows) shows the direction the flow (V, w) would take if the system started at each point.

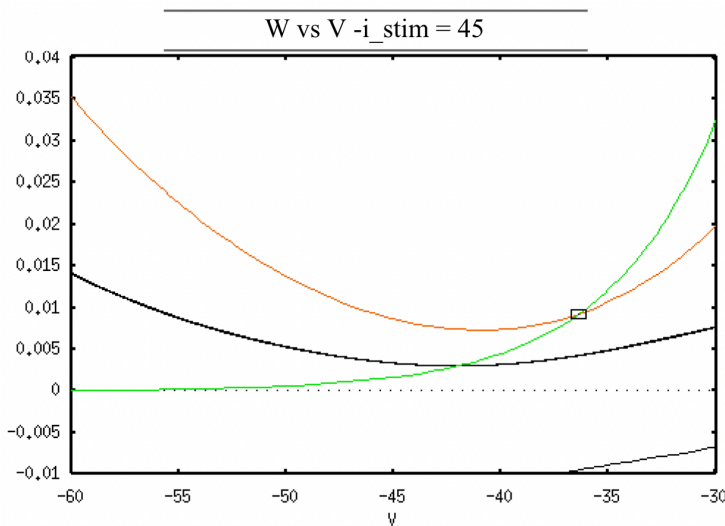
At approximately $V = -38$ and $w=0.01$ the V- and w-nullclines collide, indicating the presence of a fixed point. This appears to cause the stable fixed point to be destroyed, pushing the system into instability and into a limit cycle causing excitation. This is observed by the trajectory spiral around the fixed point, along with the direction field being centred around it. If this fixed point is indeed unstable, pushing the system into a limit cycle, this can be validated by evaluating the Jacobian eigenvalues. A positive eigenvalue will indicate instability at the fixed point and prove that the system has undergone a Hopf bifurcation and moved into a limit cycle. This indicates that the neuron has therefore moved into a state of excitation causing the system to create repetitive spiking.



← At $I_{stim} = 0$, the V-nullcline still intersects with the w-nullcline, but the fixed point is stable, as indicated by inward pointing trajectories in the direction field and negative eigenvalues. This shows that the system always remains stable and is never pushed into a limit cycle while $I_{stim} = 0$, and therefore never exhibits repetitive spiking behaviour.

Now let's analyse how our phase planes change at the point just before excitation and just after. We know from our time series analysis that the point just before excitation for $\beta_w = -13$ is 42, and the system is pushed into excitation at $I_{stim} = 43$. So let's compare phase planes and eigenvalues at these various points:





Close	STABLE	Import
c+ = 0	c- = 2	im = 0
r+ = 0	r- = 0	
V=-39.468		
W=0.0049984		

Close	UNSTABLE	Import
c+ = 2	c- = 0	im = 0
r+ = 0	r- = 0	
V=-36.385		
W=0.0092214		

Close	UNSTABLE	Import
c+ = 2	c- = 0	im = 0
r+ = 0	r- = 0	
V=-38.315		
W=0.0062871		

Eigenvalue for $i_{stim} = 42$ is **stable** -> no limit cycle
 Eigenvalue for $i_{stim} = 43$ is **unstable** -> limit cycle
 Eigenvalue for $i_{stim} = 45$ is **unstable** -> limit cycle

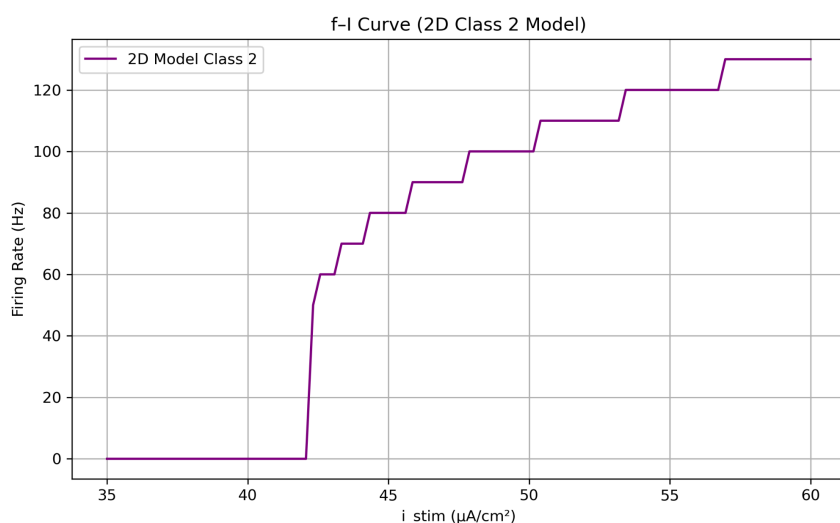
For $I_{stim} = 42$, it appears as though the system enters a self-limiting spiral around the fixed point which settles into a stable state. The implication here being that the system remains stable, does not enter a limit cycle, and therefore does not generate repetitive spiking after the initial transient spike.

However for $I_{stim} = 43$ and $I_{stim} = 45$ it appears as though the trajectories spiral away from the fixed point where the v- and w-nullclines intersect, forming a limit cycle. The implication here being that the system has been pushed into an unstable state, and entered a non-limiting oscillatory loop, producing repetitive spiking in the neuron.

Indeed our visual analysis of the phase plane plots is confirmed by calculating the eigenvalues and their stability at the points just before excitation ($I_{stim}=42$) and just after ($I_{stim}=43$ and 45).

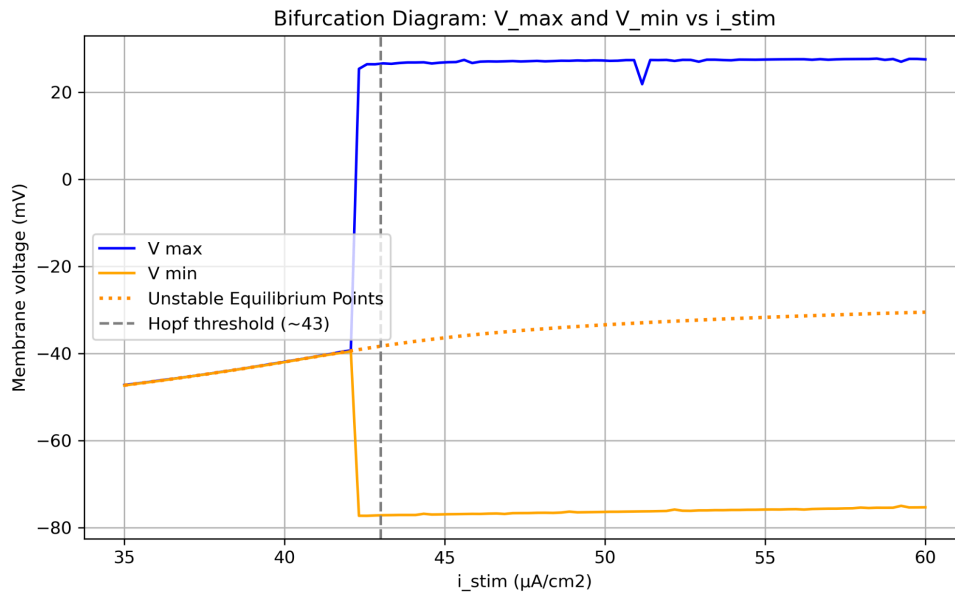
The visual phase-plane analysis and eigenvalue analysis all confirm behavior consistent with a subcritical Hopf bifurcation, where the resting state loses stability and the system jumps into a limit cycle. This validates what is proposed in the Prescott study as the mechanism from which class 2 excitability is displayed in spinal lamina I neurons.

These findings are further validated by the f-I curves for $\beta_w = -13$:



← As is visible from the f-I plot, the system produces 0 spikes at I_{stim} values up to 42, after which point the system leaps into a relatively high firing rate of 60Hz, indicating a sudden jump into a limit cycle. This is again indicative of a class 2 neuron that is unable to sustain low firing rates at low stimulation values.

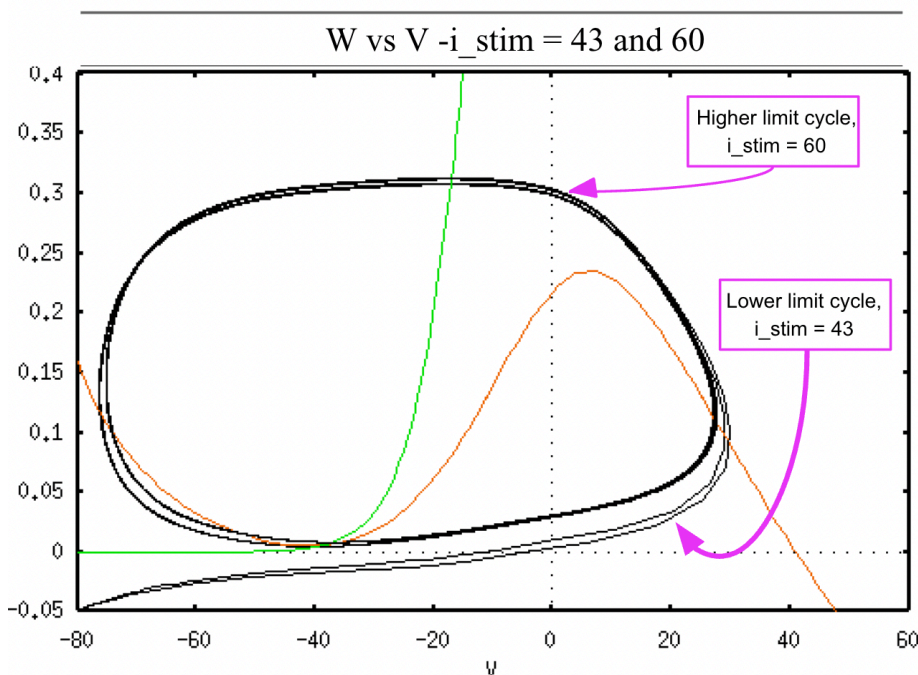
Bifurcation Analysis



Our bifurcation plot above shows the maximum and minimum membrane voltage as I_{stim} is increased. We can see that from I_{stim} 0 to 42, both V_{max} and V_{min} are equivalent, indicating a steady membrane voltage and therefore no spiking at subthreshold I_{stim} levels. At $I_{stim} = 43$ we see the system quickly jump into V_{max} and V_{min} widely diverging, indicating that the membrane voltage has started oscillating between approx -78mV and +25mV - showing a spike size of approx 103mV. This plot again validates our previous time series and phase plane analysis and shows that at $I_{stim} = 43$ the system has undergone a Hopf bifurcation, entered a limit cycle, become unstable, and produced repetitive spiking.

The dotted orange line that shows the unstable equilibrium points indicates that, as I_{stim} increases, the fixed points (i.e. where the v- and w-nullclines intersect) also increase, moving the limit cycle upward, and therefore increasing the firing rate.

This is also evidenced by viewing the phase planes for $I_{stim} = 43$ and $I_{stim} = 60$ below:



The $I_{stim} = 43$ limit cycle is lower and wider showing that the neuron spends more time cycling (i.e. a higher time recovering before spiking again)

The $I_{stim} = 60$ limit cycle is higher and narrower showing that the limit cycle now spends less time recovering and is able to fire a new spike quicker, thus increasing the firing rate.

References

For .ode files used to generate plots in XXPAUT and python code used to generate bifurcation, f-I, and heatmap plots <https://github.com/CharlotteStevensBristol/MathBiology-Neuron-Project>

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AI usage statement - I acknowledge the use of an AI grammar checker via Google Sheets to proof-read my own work and correct grammatical errors.